SDG BLOCKCHAIN ACCELERATOR

Technical Architecture Document

SolarVillage: A Hydra-based Energy Settlement & Carbon Issuance Architecture

1. Project Information

- **Project Name:** SolarVillage
- **Challenge & UNDP Office:** UNDP Nigeria
- **Document Version:** v1.0 (Dec 23, 2025)

2. Overview

SolarVillage is a decentralized energy settlement and carbon accounting platform designed for rural and peri-urban microgrids, starting with pilots in Nigeria.

The project addresses three interrelated problems:

1. **Energy access and payment reliability** in communities with limited banking penetration.
2. **Operational transparency and auditability** for microgrid operators, investors, and regulators.
3. **Credible carbon credit generation** from clean energy delivery, with reduced fraud and double-counting risk.

The system uses **Cardano Hydra** for high-frequency, low-cost operational accounting (metering, tariffs, settlements) and **Cardano Layer 1 (L1)** for low-frequency, high-assurance issuance of carbon assets.

Key features include:

- Custodial consumer balances (no consumer wallets required)
- Real-time settlement in an internal NGN-denominated accounting credit (SVN)
- Deterministic accumulation of delivered energy (WH) for audit and carbon eligibility
- Periodic issuance of **fungible carbon credits (grams CO_2e)** and a **certificate NFT** on Cardano L1
- Alignment with **Verra / Gold Standard–style methodologies** and the forthcoming **Nigerian Carbon Market (NCCC) registry**

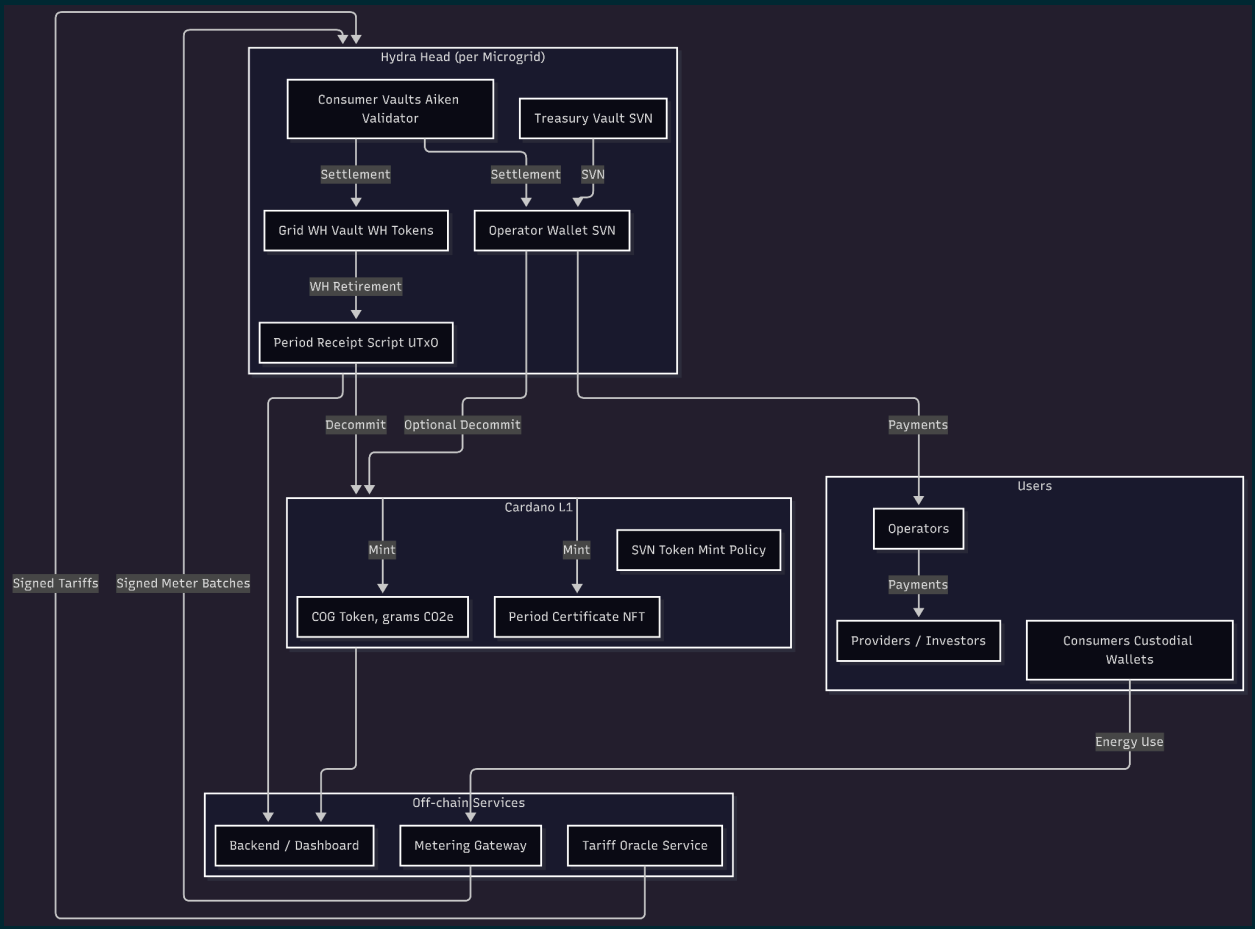
Smart contract logic is implemented primarily in **Aiken**, both for Hydra-side validators and L1 minting policies, with off-chain services coordinating oracle inputs, signatures, and anchoring.

3. System Architecture Diagram

The SolarVillage system is split into two tightly coupled layers:

- **Hydra Layer (Operational Ledger):**
  - High-frequency settlement
  - Custodial user balances
  - Tariff application
  - Energy (WH) accounting
  - Period receipt generation
- **Cardano L1 (Issuance & Registry Ledger):**
  - SVN minting
  - Carbon credit (COG) minting
  - Certificate NFT issuance
  - Long-term audit and provenance

3.1 High-level architecture



3.2 Component roles

User wallets

- Consumers: custodial, non-interactive
- Operators / providers / investors:
  - Bring-your-own wallet **or**
  - App-provided non-custodial wallet

Aiken validator scripts

- Consumer vault control

- Settlement logic
- WH minting and retirement
- Period receipt creation
- L1 minting policies for COG + NFT

Off-chain components

- Metering gateway (signed readings)
- Tariff oracle service
- Backend indexing (Hydra + L1)
- User dashboards and reporting

Cardano Ledger (UTxO)

- Immutable audit trail
- Asset issuance and provenance

Oracles / external data

- Meter readings
- Tariff updates
- Carbon methodology parameters

4. Blockchain Design

SolarVillage uses a **UTxO-native design** across both Hydra and Cardano L1. Smart contracts are written primarily in **Aiken**, with Hydra Head infrastructure used for execution and coordination.

4.1 Smart Contracts

4.1.1 Validator scripts (Hydra)

The following Aiken validators are deployed inside each Hydra head (one head per microgrid):

1. **Consumer Vault Validator**
    - Purpose: manage custodial consumer balances
    - Prevents consumers from blocking debits
    - Allows only authorized settlement transitions
2. **Settlement Validator**
    - Purpose: apply metering + tariff logic
    - Atomically:
        - debits consumer balances (SVN),
        - credits operator revenue (SVN),
        - mints WH tokens to the grid entity
3. **Grid WH Vault Validator**
    - Purpose: hold and protect WH tokens
    - Ensures WH can only be:
        - minted via settlement, or
        - retired via period close
4. **Period Receipt Validator**
    - Purpose: create a single receipt per `(grid_id, period_id)`

- Commits to all facts required for carbon issuance
- Produces a script-controlled UTxO later decommitted to L1

All validators are written in Aiken and executed inside the Hydra head.

4.1.2 Minting policies (Cardano L1)

On L1, Aiken minting policies govern:

1. **SVN minting**

    - Minted only by the platform operator
    - Backed by off-chain fiat or cNGN custody
    - SVN is an **internal accounting credit**, not a public stablecoin

2. **COG (Carbon Offset Grams) minting**

    - Minted only when a valid Period Receipt UTxO is consumed
    - Unit: grams CO_2e
    - Quantity derived deterministically from retired WH and methodology parameters

3. **Period Certificate NFT minting**

    - One NFT per `(grid_id, period_id)`
    - Minted atomically with COG
    - Acts as the public issuance certificate

4.2 Datum structures (conceptual)

4.2.1 Consumer Vault Datum

```
{
  consumer_id,
  grid_id,
  svn_balance,
  last_settlement_timestamp
}
```

4.2.2 Grid WH Vault Datum

```
{
  grid_id,
  wh_balance
}
```

4.2.3 Period Receipt Datum

```
{
  grid_id,
  period_id,
  wh_retired_total,
  methodology_id,
  period_root,
  signer_set
}
```

The `period_root` is a hash committing to:

- all interval settlements,
- tariff history,
- metering batch roots,
- methodology parameters.

4.3 Redeemer structures

Redeemers are used to restrict allowed actions:

```
SettlementRedeemer = ApplyInterval
WhRedeemer         = Mint | Retire
ReceiptRedeemer    = Create
CogRedeemer        = Issue
```

Each validator explicitly pattern-matches the redeemer and rejects all others.

4.4 UTxO model usage

SolarVillage relies on **explicit state transitions** rather than implicit balances.

- Inputs:
    - consumer vaults
    - treasury vault
    - tariff reference
    - metering batch commitment
- Outputs:
    - updated vault UTxOs
    - operator SVN UTxOs
    - grid WH vault updates

This ensures:

- deterministic replay,
- full auditability,
- no hidden state.

4.5 Token management

4.5.1 SVN (internal accounting credit)

- Minted on L1
- Committed into Hydra
- Freely transferable inside Hydra
- Decommittable back to L1
- Redeemable off-chain by KYC'd participants

4.5.2 WH (audit & carbon eligibility)

- Minted **only inside Hydra**
- Unit = 1 watt-hour
- Non-monetary
- Accumulates to grid entity
- Retired before carbon issuance

4.5.3 COG (carbon credits)

- Minted on L1
- Unit = grams CO_2e
- Backed by retired WH
- Fungible and tradable

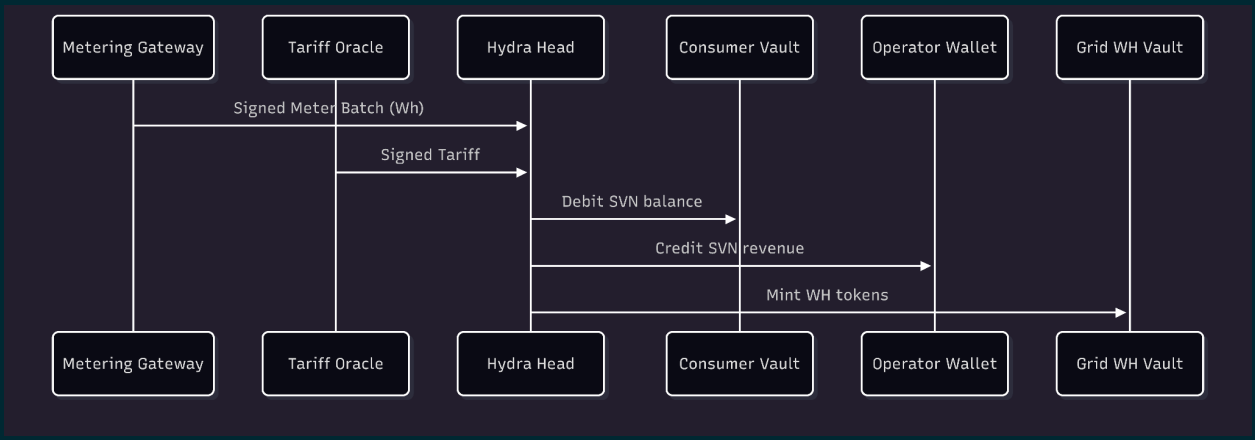
4.6 Security considerations

- **Signature checks**
    - Operator gateway signs tariffs and metering batches
    - Grid entity co-signs carbon issuance
- **Datum validation**
    - All balances and quantities must reconcile
- **Replay protection**
    - Period Receipt is consumed during issuance
- **Double counting prevention**
    - WH must be retired before issuance
    - Receipt is single-use

5. Data Flow & Transaction Lifecycle

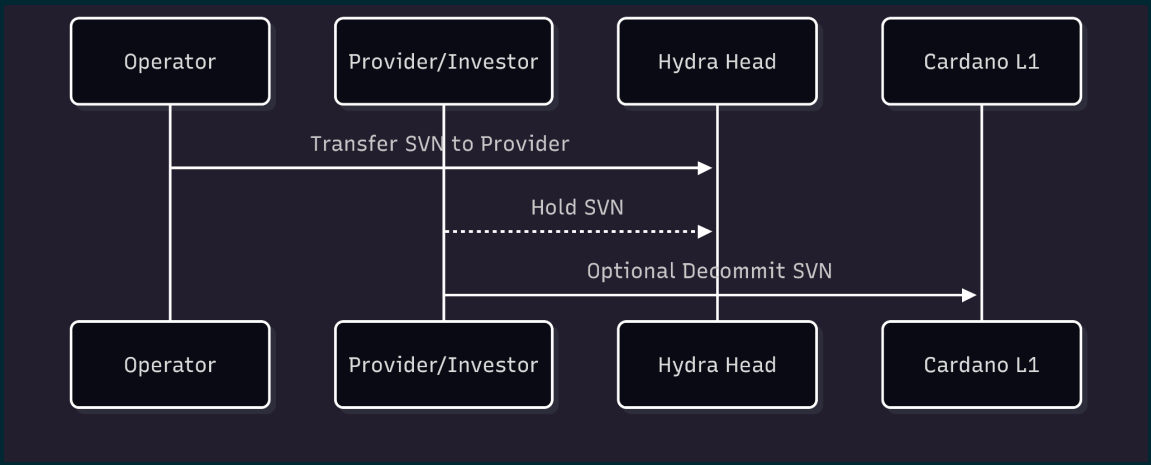
This section describes how data moves through the system from energy usage to carbon issuance.

5.1 Settlement lifecycle (15-minute intervals)



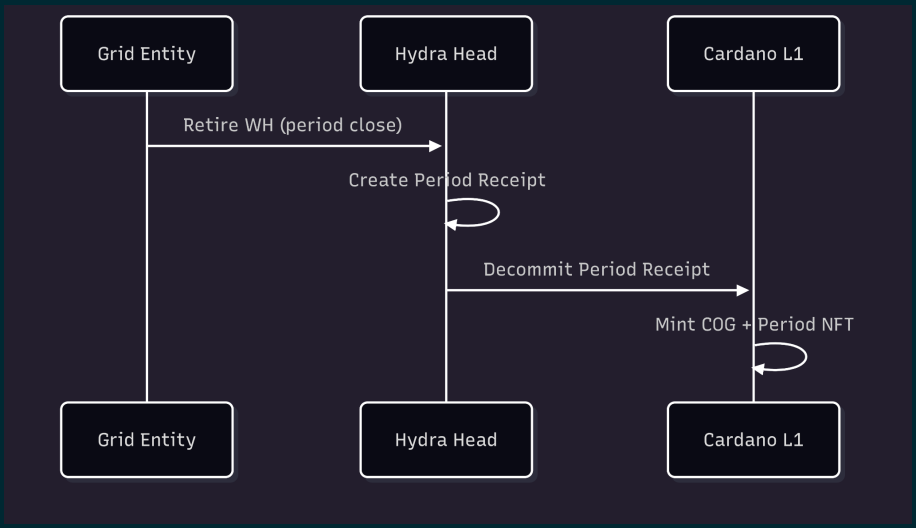
All steps occur in a **single Hydra transaction**.

5.2 Payment lifecycle (SVN)



Decommit is required **only** when leaving Hydra.

5.3 Carbon lifecycle (periodic)



6. Off-chain Components

Off-chain services coordinate data ingestion, signatures, and user visibility. They **do not** have discretionary control over on-chain outcomes.

6.1 Backend services

- Aggregates metering data
- Prepares signed meter batches
- Publishes tariff updates
- Submits Hydra transactions

6.2 Cardano integration

- Cardano-node (L1)
- Hydra-node per grid
- Aiken for script compilation
- Ogmios / Kupo for indexing and monitoring

6.3 Oracles and data providers

- Metering devices → gateway
- Tariff management service
- Carbon methodology parameters (off-chain config, hashed on-chain)

6.4 Dashboards and reporting

- Consumer energy usage
- Operator revenue (SVN)
- Grid energy delivery (WH)
- Carbon issuance status (COG + NFT)

Dashboards are **derived views**, not sources of truth.

7. Sandbox/Testnet Results TBD

- Include logs, screenshots, or transaction IDs demonstrating successful test deployments.
- Mention metrics like execution units, memory consumption, or transaction success rates.

Example:

Transaction ID	Type	Status	CPU	Memory	Notes
abc123	Mint	Success	310k	900k	Validator passed all checks
def456	Transfer	Success	280k	850k	Tested edge case: invalid redeemer

8. Tools and Environments Used

SolarVillage is built using a combination of Cardano-native tooling, Hydra infrastructure, and standard off-chain services. The toolchain is selected to maximize correctness, auditability, and long-term maintainability.

8.1 Blockchain infrastructure

- **Cardano Node**
    - Used for L1 asset minting, anchoring, and long-term audit
    - Deployment targets:
        - Preprod (initial testing)

        - Mainnet (production)

- **Hydra Head**

    - One long-lived head per microgrid
    - Handles high-frequency settlement and accounting
    - Supports incremental decommit of selected UTxOs (SVN, Period Receipts)

8.2 Smart contract development

- **Aiken**

    - Primary smart contract language for:

        - Hydra validators (consumer vaults, settlement, WH vaults, period receipts)
        - L1 minting policies (SVN, COG, Period Certificate NFT)

    - Used for:

        - deterministic pattern matching on redeemers
        - strict datum validation
        - minimizing script size and execution costs

Where appropriate (e.g. Hydra node internals), non-Aiken components rely on:

- Haskell-based Hydra infrastructure
- Standard Cardano ledger semantics

8.3 Off-chain services and integration

- **Hydra Node**

    - Manages head lifecycle and transaction submission
    - Interfaces with off-chain services

- **Ogmios**

    - JSON-RPC interface for Cardano node
    - Used for transaction submission and chain queries

- **Kupo**
    - Lightweight indexer for UTxO tracking
    - Used for monitoring receipts, issuance events, and balances
- **Backend services**
    - Metering aggregation
    - Tariff management
    - Signature orchestration
    - User dashboards and reporting

These services **coordinate**, but do not control, on-chain outcomes.

8.4 Environments

- **Local / emulator**
    - Aiken test framework
    - Hydra devnet for early validation
- **Cardano Preprod**
    - End-to-end testing of:
        - decommit flows
        - minting policies
        - NFT metadata
        - anchor transactions
- **Production**
    - Cardano Mainnet
    - Dedicated Hydra heads per grid

9. Remaining Considerations / Next Steps

The current architecture is intentionally conservative and extensible. The following areas represent planned or optional future work.

9.1 Performance and scalability

- Stress testing Hydra heads with:
    - high meter counts
    - burst settlement scenarios
- Optimizing Aiken scripts to:
    - reduce execution units
    - minimize datum size
- Exploring batching strategies for:
    - settlement intervals
    - period receipt creation

9.2 Security and audit readiness

- Independent review of:
    - settlement validator invariants
    - receipt consumption logic
    - minting policy correctness
- Formal verification of:
    - WH → COG conversion constraints
    - double-issuance prevention
- Documentation package for:
    - UNDP review
    - third-party auditors
    - carbon registry assessors

9.3 Carbon registry integration

- Mapping Period Certificate NFTs to:
  - Verra / Gold Standard registry records
  - forthcoming **Nigerian Carbon Market (NCCC)** identifiers
- Supporting:
  - off-chain attestations
  - registry-side retirement acknowledgements
- Optional future bridge:
  - on-chain COG ↔ registry-issued credits

9.4 Governance and decentralization

- Transitioning grid entities to:
  - on-chain DAOs
  - multi-signature control over WH retirement and issuance
- Gradual reduction of platform operator privileges
- Transparent role separation between:
  - platform
  - operators
  - grid entities

9.5 Regulatory and compliance considerations

- Explicit positioning of SVN as:
  - an **internal accounting credit**
  - available only to KYC'd participants
- Ongoing review of:
  - Nigerian financial regulations

    - carbon market compliance requirements

- Clear disclosure and reporting standards for:
    - users
    - investors
    - regulators

Appendix A: Carbon Issuance & Registry Alignment

Below is a **stand-alone Carbon Issuance Appendix** extracted from the above architecture and rewritten specifically for **carbon registry engagement**. It is suitable to attach to submissions or discussions with **Verra**, **Gold Standard**, and the forthcoming **Nigerian Carbon Market** (sometimes also referenced as the NCCC framework).

A.1 Purpose of this appendix

This appendix explains how SolarVillage:

- generates carbon-credit-eligible energy data,
- prevents double counting and replay,
- issues carbon credits on-chain in a registry-aligned manner,
- and supports verification, attestation, and future registry integration.

The appendix is written for **registry operators, auditors, and MRV reviewers**, not blockchain engineers.

A.2 Carbon asset model (high level)

SolarVillage separates **energy delivery**, **carbon eligibility**, and **carbon issuance** into distinct layers:

Layer	Representation	Purpose
Energy delivery	Metered watt-hours (Wh)	Physical reality
Carbon eligibility	WH audit tokens (Hydra)	Prevent double counting
Carbon issuance	COG (grams CO_2e) + Certificate NFT (L1)	Market-facing credits

Only **retired energy** can generate carbon credits.

A.3 Measurement, Reporting, and Verification (MRV)

A.3.1 Measurement

- Electricity delivery is measured at **15-minute intervals** per meter.
- Meter readings are aggregated by a **metering gateway** and cryptographically signed.
- Measurements are immutable once recorded inside the Hydra ledger.

A.3.2 Reporting

- All interval settlements are recorded as UTxO state transitions.
- Each settlement commits to:
    - meter batch hash,
    - tariff reference,
    - timestamp window.
- These records form a **complete, replayable audit trail**.

A.3.3 Verification

An independent verifier can:

1. Recompute delivered Wh from metering data.
2. Verify tariff application.
3. Verify aggregation into issuance periods.
4. Verify that retired Wh equals issued carbon credits.

No trust in the platform operator is required.

A.4 Energy-to-carbon conversion logic

A.4.1 Canonical unit

- Carbon credits are issued in **grams of CO_2e (COG)**.
- This avoids rounding errors and supports fine-grained accounting.

A.4.2 Conversion formula

For each issuance period:

```
COG_grams = WH_retired × Methodology_Factor
```

Where:

- `WH_retired` = total watt-hours retired for the period
- `Methodology_Factor` = grams CO_2e avoided per Wh, defined by the selected methodology

The methodology factor is:

- versioned,
- referenced immutably during issuance,
- auditable independently.

A.5 Prevention of double counting

SolarVillage enforces **three independent safeguards**:

A.5.1 Physical layer

- Each meter reading is uniquely identified by `(meter_id, timestamp_window)`.

A.5.2 Accounting layer (Hydra)

- WH tokens are **retired (burned or locked)** before carbon issuance.
- Retired WH cannot be reused in future periods.

A.5.3 Issuance layer (L1)

- Each issuance requires consuming a **Period Receipt**.
- A Period Receipt is **single-use** and destroyed during issuance.

This prevents:

- duplicate issuance,
- replay attacks,
- retroactive manipulation.

A.6 Period-based issuance model

A.6.1 Issuance periods

- Issuance occurs at fixed periods (e.g. weekly or monthly).
- Each period is uniquely identified by:
    - `grid_id`
    - `period_id`

A.6.2 Period Receipt

At period close:

- All eligible WH is retired.
- A **Period Receipt** is created with a cryptographic commitment to:
    - total retired Wh,
    - all included settlements,
    - tariff history,
    - methodology version.

This receipt acts as a **digital equivalent of a registry issuance request**.

A.7 On-chain issuance artifacts

A.7.1 Fungible carbon credits (COG)

- Native Cardano assets.
- Unit: grams CO_2e.
- Fully fungible within the same methodology and vintage.
- Minted only when a valid Period Receipt is consumed.

A.7.2 Period Certificate NFT

- One NFT per `(grid_id, period_id)`.
- Represents the **issuance certificate**, not the tradable unit.

- Metadata includes:
    - retired Wh total,
    - issued COG amount,
    - methodology identifier,
    - cryptographic roots of supporting data.

This mirrors how registries separate:

- credit units from
- issuance documentation.

A.8 Alignment with existing registries

A.8.1 Verra / Gold Standard alignment

SolarVillage mirrors standard registry concepts:

Registry concept	SolarVillage equivalent
Monitoring report	Interval settlement records
Issuance batch	Period Receipt
Credit units	COG (grams CO_2e)
Retirement	Receipt consumption
Certificate	Period Certificate NFT

This enables:

- straightforward attestation,
- registry-side mirroring or bridging,
- hybrid on-chain/off-chain issuance models.

A.8.2 Nigerian Carbon Market (NCMC / NCCC)

The architecture is designed to:

- reference national methodologies,
- include Nigerian grid identifiers,
- support sovereign oversight and reporting requirements.

The Period Certificate NFT can act as:

- a cryptographic anchor for registry entries, or
- a mirror of registry-issued credits.

A.9 Audit and transparency guarantees

Every issued carbon credit can be traced:

COG (grams CO_2e)

→ Period Certificate NFT

→ Period Receipt (consumed)

→ Retired WH

→ Interval settlements

→ Meter readings + tariffs

This traceability is:

- immutable,
- machine-verifiable,
- regulator-friendly.

A.10 Intended registry interaction models

SolarVillage supports multiple engagement paths:

1. **On-chain-first, registry-mirrored**

    - Credits issued on-chain, mirrored into a registry.

2. **Registry-first, on-chain-represented**
    - Registry issues credits; on-chain assets represent them.
3. **Hybrid attestation**
    - Registry validates Period Certificates and attests externally.

The architecture does **not** lock SolarVillage into a single model.

A.11 Summary for registry reviewers

- No carbon credit can be issued without **retiring real, metered energy**.
- No retired energy can be used more than once.
- Issuance is **periodic, deterministic, and auditable**.
- On-chain artifacts map cleanly to registry concepts.
- The system is designed for **regulatory compatibility, not circumvention**.

Appendix 2: Mermaid Diagram Source Codes

Diagram Source code: Section 3.1 High-level architecture

```
---
config:
  theme: redux-dark
  layout: elk
  look: neo
---
flowchart TB
 subgraph Users["Users"]
        C["Consumers Custodial Wallets"]
        O["Operators"]
        P["Providers / Investors"]
  end
 subgraph Hydra["Hydra Head (per Microgrid)"]
        CV["Consumer Vaults Aiken Validator"]
        TV["Treasury Vault SVN"]
        OW["Operator Wallet SVN"]
        GW["Grid WH Vault WH Tokens"]
        PR["Period Receipt Script UTxO"]
  end
 subgraph OffChain["Off-chain Services"]
        M["Metering Gateway"]
        T["Tariff Oracle Service"]
        B["Backend / Dashboard"]
  end
 subgraph L1["Cardano L1"]
        SVN["SVN Token Mint Policy"]
        COG["COG Token, grams CO2e"]
        NFT["Period Certificate NFT"]
  end
    C -- Energy Use --> M
    M -- Signed Meter Batches --> Hydra
    T -- Signed Tariffs --> Hydra
    CV -- Settlement --> OW & GW
    TV -- SVN --> OW
```

```
    GW -- WH Retirement --> PR
    PR -- Decommit --> L1
    L1 -- Mint --> COG & NFT
    OW -- Payments --> O
    O -- Payments --> P
    OW -- Optional Decommit --> L1
    Hydra --> B
    L1 --> B
```

Diagram Source code: Section 5.1 Settlement lifecycle (15-minute intervals)

```
sequenceDiagram

  participant Meter as Metering Gateway

  participant Oracle as Tariff Oracle

  participant Hydra as Hydra Head

  participant Vault as Consumer Vault

  participant Op as Operator Wallet

  participant Grid as Grid WH Vault

  Meter->>Hydra: Signed Meter Batch (Wh)

  Oracle->>Hydra: Signed Tariff

  Hydra->>Vault: Debit SVN balance

  Hydra->>Op: Credit SVN revenue

  Hydra->>Grid: Mint WH tokens
```

Diagram Source code: Section 5.2 Payment lifecycle (SVN)

```
sequenceDiagram

    participant Op as Operator

    participant Prov as Provider/Investor

    participant Hydra as Hydra Head

    participant L1 as Cardano L1

    Op->>Hydra: Transfer SVN to Provider

    Prov-->>Hydra: Hold SVN

    Prov->>L1: Optional Decommit SVN
```

Diagram Source code: Section 5.3 Carbon lifecycle (periodic)

```
sequenceDiagram

    participant Grid as Grid Entity

    participant Hydra as Hydra Head

    participant L1 as Cardano L1

    Grid->>Hydra: Retire WH (period close)

    Hydra->>Hydra: Create Period Receipt

    Hydra->>L1: Decommit Period Receipt
```

```
    L1->>L1: Mint COG + Period NFT
```